

Evaluation of Quality of Endodontic Re-Treatment and Changes in Periapical Status

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INTRODUCTION

Apical periodontitis (AP) may result from dental caries, fracture tooth, traumatic occlusion, etc. The main treatment for AP is root canal treatment (RCT).^[1] It is evident from numerous studies that the quality of endodontic treatment performed in general practice is less superior than those performed in specialized dentistry.^[2] Hence, the success rate of RCT is obviously high. In spite of better treatment outcome, failure rate cannot be completely avoided. The presence of tenderness in RC treated tooth and radiological evidence of periodontal ligament widening and loss of lamina dura is indicative of failed RCT.^[3]

There is a high prevalence of AP that ranged from 8%–72%. Numerous studies have done so far depicting prevalence of AP using various intraoral radiographs such as intraoral peri-apical radiographs (IOPAR) and panoramic radiographs. These radiographic aids helped in assessing the periapical region as well as in detecting quality of the RCT.^[4,5]

Recently, newer radiographic diagnostic aid such as cone-beam computed tomography (CBCT) has emerged with better image quality as compared to intraoral radiographs such as IOPARs. AN accuracy of 0.54 with panoramic radiographs and 0.70 with peri-apical radiographs has been found. The only drawback of CBCT is high radiation exposure as compared to single intraoral peri-apical radiograph.^[6] The present study was conducted to assess the quality of RC filling before and after RC re-treatment.

MATERIALS AND METHODS

This retrospective study was initiated in the department of endodontics which comprised of 238 radiographs of failed endodontic treatment. Ethical committee of the institute was approached for the approval and after explaining the utility of the study approval was taken.

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Radiographs which were of poor quality, nondiagnostic, and with radiographic errors were not considered to be involved in the study. Digital IOPARs were taken with Schick sensor using Gnatus intraoral radiographic unit operating at 72 kVp, 8mA tube current and exposure time ranged from 0.30–0.50 sec. Rinn X tension C-one P-parallelizing (XCP) holder was used for holding sensor. The periapical status of the endodontic treatment was evaluated with periapical index (PAI) scoring system. Scoring ranged from 1 to 5 was opted. The PAI scores were based on absence and presence of periapical lesion where score 0 was indicative of absence of pathology and 1 suggested presence of pathology. PAI <3 showed absence and PAI >3 showed presence of periapical lesion.

Baseline and follow-up radiographs at 6 months were compared. Density of the filling and the distance between the end of the filling and the radiographic apex indicated the quality of RC filling that scored from 1 to 6 based on scoring suggested in study by Unal *et al.*^[6] The absence of voids and the condensation of the filling material in the RC indicated of homogenous RC filling. A RCT with an acceptable filling length and a homogenous root filling was defined as being an adequate RCT.

emales were 40.5 years. Almamoudi and Al-Saleh¹⁴ There are no conflicts of interest.

REFERENCES

1. Barrieshi-Nusair KM, Al-Omari MA, Al-Hiyasat AS. Radiographic technical quality of root canal treatment performed by dental students at the Dental Teaching Center in Jordan. *J Dent* 2004;32:301-7.
2. Fezai H, Al-Salehi S. The relationship between endodontic case complexity and treatment outcomes. *J Dent* 2019;85:88-92.
3. Bierenkrant DE, Parashos P, Messer HH. The technical quality of nonsurgical root canal treatment performed by a selected cohort of Australian endodontists. *Int Endod J* 2008;41:561-70.
4. Siqueira JF Jr. Aetiology of root canal treatment failure: Why well-treated teeth can fail. *Int Endod J* 2001;34:1-10.
5. Covello F, Franco V, Schiavetti R, Clementini M, Mannocci A, Ottria L, *et al.* Prevalence of apical periodontitis and quality of endodontic treatment in an Italian adult population. *Oral Implantol* 2010;3:9-14.
6. Unal GC, Kececi AD, Kaya BU, Tac AG. Quality of root canal fillings performed by undergraduate dental students. *Eur J Dent* 2011;5:324-30.
7. Chala S, Abouqal R, Abdallaoui F. Prevalence of apical periodontitis and factors associated with the periradicular status. *Acta Odontol Scand* 2011;69:355-9.
8. Abbott PV. Recognition and prevention of failures in clinical dentistry. *Endodontics. Ann R Australas Coll Dent Surg* 1991;11:150-66.
9. Hession RW. Long-term evaluation of endodontic treatment: Anatomy, instrumentation, obturation – The endodontic practice triad. *Int Endod J* 1981;14:179-84.
10. Alharmoodi R, Al-Salehi S. Assessment of the quality of endodontic re-treatment and changes in periapical status on a postgraduate endodontic clinic. *J Dent* 2019;92:103261.
11. Ng YL, Gulabivala K. Outcome of non-surgical re-treatment. *Endod Top* 2008;18:3-30.
12. Lambrianidis T, Tosounidou E, Tzoanopoulou M. The effect of maintaining apical patency on periapical extrusion. *J Endod* 2001;27:696-8.